LAB 7

#1

```
#include<iostream>
using namespace std;

class polygon
{
protected:
    int width,height;

public:
    void set_value(int w,int h)
    {
        width=w;
        height=h;
    }
};

class rectangle: public polygon
{
public:
    int calculate_area()
    {
        return width*height;
    }
};

class triangle: public polygon
{
public:
    int calculate_area()
    {
        return width*height/2;
    }
};

int main()
{
    polygon *p;
    rectangle r;
    triangle t;

    p=&r;
    p->set_value(10,5);
    cout<<"Rectangle area: "<<r.calculate_area()<<endl;

    p=&t;
    p->set_value(10,5);
    cout<<"Triangle area: "<<t.calculate_area()<<endl;
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
Rectangle area: 50
Triangle area: 25
```

#2

```
#include<iostream>
using namespace std;

class shape
{
public:
    virtual void area(){}
    virtual void display(){}
};

class circle: public shape
{
    float r;
public:
    circle(float x){r=x;}

    void area()
    {
        cout<<"Area of circle: "<<3.14*r*r<<endl;
    }

    void display()
    {
        cout<<"Shape: Circle"<<endl;
    }
};

class rectangle: public shape
{
    float l,b;
public:
    rectangle(float x,float y){l=x;b=y;}

    void area()
    {
        cout<<"Area of rectangle: "<<l*b<<endl;
    }

    void display()
    {
        cout<<"Shape: Rectangle"<<endl;
    }
};

class triangle: public shape
{
    float b,h;
public:
    triangle(float x,float y){b=x;h=y;}

    void area()
    {
        cout<<"Area of triangle: "<<0.5*b*h<<endl;
    }

    void display()
```

```
    {
        cout<<"Shape: Triangle"<<endl;
    }
};

int main()
{
    shape *s;

    circle c(5);
    rectangle r(4,6);
    triangle t(3,7);

    s=&c; s->display(); s->area();
    s=&r; s->display(); s->area();
    s=&t; s->display(); s->area();
}
```

```
Shape: Circle
Area of circle: 78.5
Shape: Rectangle
Area of rectangle: 24
Shape: Triangle
Area of triangle: 10.5
PS C:\Users\lenovo\Desktop\DSA>
```

#3

```
#include<iostream>
#include<cmath>
using namespace std;

float area(float b,float h)
{
    return 0.5*b*h;
}

float area(float a)
{
    return (1.732/4)*a*a;
}

float area(float a,float b,float c)
{
    float s=(a+b+c)/2;
    return sqrt(s*(s-a)*(s-b)*(s-c));
}

int main()
{
    cout<<"Right triangle: "<<area(4,5)<<endl;
    cout<<"Equilateral: "<<area(6)<<endl;
    cout<<"Isosceles: "<<area(5,5,6)<<endl;
}
```

```
Right triangle: 10
Equilateral: 15.588
Isosceles: 12
```

#4

```
#include<iostream>
using namespace std;

class Student
{
public:
    virtual void show()=0;
};

class Engineering: public Student
{
public:
    void show()
    {
        cout<<"Engineering student"<<endl;
    }
};

class Medicine: public Student
{
public:
    void show()
    {
        cout<<"Medicine student"<<endl;
    }
};

class Science: public Student
{
public:
    void show()
    {
        cout<<"Science student"<<endl;
    }
};

int main()
{
    Student *s[3];

    Engineering e;
    Medicine m;
    Science sc;

    s[0]=&e;
    s[1]=&m;
    s[2]=&sc;

    for(int i=0;i<3;i++)
        s[i]->show();
```

```
}
```

```
Engineering student
Medicine student
Science student
```

#5

```
#include<iostream>
using namespace std;

class Time
{
    int h,m,s;

public:
    Time(){}
    Time(int a,int b,int c)
    {
        h=a; m=b; s=c;
    }

    Time operator+(Time t)
    {
        Time temp;
        temp.s=s+t.s;
        temp.m=m+t.m+temp.s/60;
        temp.s%=60;

        temp.h=h+t.h+temp.m/60;
        temp.m%=60;

        return temp;
    }

    void show()
    {
        cout<<h<<":"<<m<<":"<<s<<endl;
    }
};

int main()
{
    Time t1(5,15,34),t2(9,53,58),t3;
    t3=t1+t2;
    t3.show();
}
```

```
15:9:32
PS C:\Users\lenovo\Desktop\DSA>
```

#6

```
#include<iostream>
#include<string.h>
using namespace std;

class STRING
```

```
{
    char s[100];

public:
    void input()
    {
        cin>>s;
    }

    bool operator==(STRING t)
    {
        return strcmp(s,t.s)==0;
    }

    STRING operator+(STRING t)
    {
        STRING temp;
        strcpy(temp.s,s);
        strcat(temp.s,t.s);
        return temp;
    }

    void display()
    {
        cout<<s<<endl;
    }
};

int main()
{
    STRING s1,s2,s3;

    s1.input();
    s2.input();

    if(s1==s2)
        cout<<"Equal"<<endl;
    else
        cout<<"Not Equal"<<endl;

    s3=s1+s2;
    s3.display();
}
```

```
PS C:\Users\lenovo\Desktop\DSA
helloo
hellooo
Not Equal
helloohellooo
```

#7

```
#include<iostream>
using namespace std;

class Matrix
{
```

```
    int a[10][10],r,c;

public:
    void input()
    {
        cout<<"Enter rows and cols: ";
        cin>>r>>c;

        for(int i=0;i<r;i++)
            for(int j=0;j<c;j++)
                cin>>a[i][j];
    }

    void display()
    {
        for(int i=0;i<r;i++)
        {
            for(int j=0;j<c;j++)
                cout<<a[i][j]<<" ";
            cout<<endl;
        }
    }

    friend Matrix operator*(Matrix,Matrix);
};

Matrix operator*(Matrix m1,Matrix m2)
{
    Matrix m3;
    m3.r=m1.r;
    m3.c=m2.c;

    for(int i=0;i<m3.r;i++)
    {
        for(int j=0;j<m3.c;j++)
        {
            m3.a[i][j]=0;
            for(int k=0;k<m1.c;k++)
                m3.a[i][j]+=m1.a[i][k]*m2.a[k][j];
        }
    }
    return m3;
}

int main()
{
    Matrix m1,m2,m3;

    cout<<"Matrix 1"<<endl;
    m1.input();

    cout<<"Matrix 2"<<endl;
    m2.input();

    m3=m1*m2;

    cout<<"Result"<<endl;
```

```
    m3.display();
}
```

```
Matrix 1
Enter rows and cols: 2 2
1 0 1 0
Matrix 2
Enter rows and cols: 2 2
1 1 1 1
Result
1 1
1 1
```

#8

```
#include<iostream>
using namespace std;

class Array
{
    int a[10];

public:
    void input()
    {
        for(int i=0;i<10;i++)
            cin>>a[i];
    }

    int operator[](int index)
    {
        if(index<0 || index>=10)
        {
            cout<<"Index out of bounds"<<endl;
            return -1;
        }
        return a[index];
    }
};

int main()
{
    Array arr;
    arr.input();
    int i;
    cout<<"Enter index: ";
    cin>>i;
    cout<<arr[i];
}
```

```
1 2 3 4 5 6 8 9 0 7
Enter index: 4
5
PS C:\Users\lenovo\Desktop\
1 2 3 4 5 6 7 8 9 0
Enter index: 11
Index out of bounds
-1
```

#9

```
#include<iostream>
using namespace std;

class Test
{
public:
    void operator()(int a,int b,int c)
    {
        cout<<"Sum = "<<a+b+c<<endl;
    }
};

int main()
{
    Test t;
    t(10,20,30);
}
```

```
PS C:\Users\lenovo\Desktop\DSA
Sum = 60
```

#10

```
#include<iostream>
using namespace std;

class Test
{
    int x;
public:
    friend istream& operator>>(istream &in,Test &t);
    friend ostream& operator<<(ostream &out,Test &t);
};

istream& operator>>(istream &in,Test &t)
{
    in>>t.x;
    return in;
}

ostream& operator<<(ostream &out,Test &t)
{
    out<<t.x;
    return out;
}
```

```
int main()
{
    Test t;
    cout<<"Enter value: ";
    cin>>t;
    cout<<"Value: "<<t;
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
Enter value: 56
Value: 56
```

#11

```
#include<iostream>
using namespace std;

class Test
{
    float x;

public:
    Test(float f)
    {
        x=f;
    }

    void show()
    {
        cout<<x;
    }
};

int main()
{
    float f;
    cout<<"Enter float: ";
    cin>>f;
    Test t=f;
    t.show();
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
Enter float: 34.5
34.5
```

#12

```
#include<iostream>
using namespace std;

class Test
{
    float x;

public:
    Test(float a)
```

```
    {
        x=a;
    }
    operator float()
    {
        return x;
    }
};

int main()
{
    Test t(5.5);
    float f=t;
    cout<<f;
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
5.5
```

#13

```
#include<iostream>
#include<cmath>
using namespace std;

class Cartesian;

class Polar
{
    float r,theta;

public:
    Polar(float a,float b)
    {
        r=a;
        theta=b;
    }

    operator Cartesian();
};
class Cartesian
{
    float x,y;

public:
    Cartesian(){}

    Cartesian(float a,float b)
    {
        x=a;
        y=b;
    }

    void show()
    {
        cout<<"x = "<<x<<" y = "<<y<<endl;
    }
};
```

```
Polar::operator Cartesian()
{
    float x=r*cos(theta);
    float y=r*sin(theta);
    return Cartesian(x,y);
}
int main()
{
    Polar p(10,5);
    Cartesian c=p;
    c.show();
}
```

```
PS C:\Users\lenovo\Desktop\DSA>
x = 2.83662 y = -9.58924
```